

PI Name: Ziyu Yao (website: <https://ziyuyao.org>; full CV: [here](#); X: [@ZiyuYao](#))

Position Title: Assistant Professor, Department of Computer Science, George Mason University

Professional Preparation:

- Ph.D. in Computer Science and Engineering, Ohio State University, 2021
- Research interns at Microsoft Semantic Machines (2021), Carnegie Mellon University (w/ Dr. Graham Neubig, 2020), Microsoft Research (2018), Fujitsu Lab of America (2017).

Related Publications: (my students are marked by*)

Mechanistic Interpretability (for Code Gen or Others):

- **[Preprint’25]** Daking Rai*, Yilun Zhou, Shi Feng, Abulhair Saparov, and **Ziyu Yao**. “A practical review of mechanistic interpretability for transformer-based language models”. Under Review, 2025. (Presented as a tutorial at ICML 2025.)
- **[NeurIPS’25]** Daking Rai*, Samuel Miller*, Kevin Moran, and **Ziyu Yao**. “**Failure by Interference: Language Models Make Balanced Parentheses Errors When Faulty Mechanisms Overshadow Sound Ones**”. To appear at NeurIPS, 2025.
- **[AAAI’25 KnowFM]** Samuel Miller*, Daking Rai*, and **Ziyu Yao**. “**Mechanistic Understanding of Language Models in Syntactic Code Completion**”. AAAI Towards Knowledgeable Foundation Models Workshop, 2025.
- **[EMNLP’25]** Siddarth Mamidanna*, Daking Rai*, **Ziyu Yao**, and Yilun Zhou. 2025. “All for One: LLMs Solve Mental Math at the Last Token With Information Transferred From Other Tokens”. In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, pages 30735–30748, Suzhou, China. Association for Computational Linguistics.
- **[EMNLP’25]** Zihao Li, Xu Wang, Yuzhe Yang, **Ziyu Yao**, Haoyi Xiong, Mengnan Du. “Feature Extraction and Steering for Enhanced Chain-of-Thought Reasoning in Language Models”. In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, pages 10904–10924, Suzhou, China. Association for Computational Linguistics.
- **[EMNLP’25 Findings]** Dong Shu, Xuansheng Wu, Haiyan Zhao, Daking Rai*, **Ziyu Yao**, Ninghao Liu, and Mengnan Du. 2025. “A Survey on Sparse Autoencoders: Interpreting the Internal Mechanisms of Large Language Models”. In Findings of the Association for Computational Linguistics: EMNLP 2025, pages 1690–1712, Suzhou, China. Association for Computational Linguistics.
- **[ACL’24]** Daking Rai* and **Ziyu Yao**. “An Investigation of Neuron Activation as a Unified Lens to Explain Chain-of-Thought Eliciting Arithmetic Reasoning of LLMs”. In Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 7174–7193. 2024.

Code Generation:

- **[Under Review’25]** Hao Yan*, Swapneel Suhas Vaidya, Xiaokuan Zhang, and **Ziyu Yao**. “Guiding AI to fix its own flaws: An empirical study on LLM-driven secure code generation”. Under Review, 2025.
- **[Under Review’25]** Hao Yan*, Thomas D. Latoza, and **Ziyu Yao**. “IntelliExplain: Enhancing Conversational Code Generation for Non-Professional Programmers.” Under Review, 2025.
- **[ACL’23]** Daking Rai*, Bailin Wang, Yilun Zhou, and **Ziyu Yao**. “Improving Generalization in Language Model-based Text-to-SQL Semantic Parsing: Two Simple Semantic Boundary-based Techniques”. In Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), pp. 150–160. 2023.
- **[ACL’23]** Hao Yan*, Saurabh Srivastava, Yintao Tai, Sida I. Wang, Wen-tau Yih, **Ziyu Yao**. “Learning to simulate natural language feedback for interactive semantic parsing”. In Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics, 2023.
- **[ICLR’21]** **Ziyu Yao**, Frank F. Xu, Pengcheng Yin, Huan Sun, and Graham Neubig. “Learning Structural Edits via Incremental Tree Transformations.” In International Conference on Learning Representations, 2021.

- **[WWW’19] Ziyu Yao**, Jayavardhan Reddy Peddamail, and Huan Sun. “Coacor: Code annotation for code retrieval with reinforcement learning.” In The World Wide Web Conference, pp. 2203-2214. 2019..
- **[WWW’18] Ziyu Yao**, Daniel S. Weld, Wei-Peng Chen, and Huan Sun. “StaQC: A systematically mined question-code dataset from Stack Overflow.” In Proceedings of the 2018 World Wide Web Conference, pp. 1693-1703. 2018.

Professional Services:

Co-editor

- ACM SIGAI Newsletter 2022 -

Organizing Committee

- **The First Workshop on the Application of LLM Explainability to Reasoning and Planning at COLM** 2025
- Diversity & Inclusion Chair at NAACL 2024
- Mid-Atlantic NLP Student Colloquium (MASC-SLL) 2023
- Workshop on Structured and Unstructured Knowledge Integration (SUKI) at NAACL 2022
- **Workshop on Natural Language Processing for Programming (NLP4Prog) at ACL** 2021

Area Chair or Senior Program Committee

- The Association for Computational Linguistics Rolling Review (ARR) 2023 -
- International Conference on Learning Representations (ICLR) 2026
- Annual Meeting of the Association for Computational Linguistics (ACL) 2023
- Empirical Methods in Natural Language Processing (EMNLP) 2022
- Annual AAAI Conference on Artificial Intelligence (AAAI) 2023, 2024

Selected Invited Talks:

- Language Model for Reasoning: The How and Why. Georgetown University, 2025.
- Mechanistically Interpreting Language Models: From Syntactic Code Completion to Arithmetic Reasoning. DaSCI Seminar at the University of Granada, 2025.
- Mechanistic Understanding of LLMs in Reasoning and Code Generation. University of Cambridge, Language Technology Lab Seminars, 2024.
- The Progress, Opportunities, and Challenges of Semantic Parsing. Bloomberg, CTO Data Science Seminar Series, 2024.
- Towards Understanding and Modeling the Interaction with Large Language Models. New York University, 2024.
- Large Language Model Cascades with Mixture of Thoughts Representations for Cost-efficient Reasoning. Microsoft Semantic Machines, 2023.
- Building Natural Language Interfaces in the Age of Large Language Models. Apple Inc., 2023.